theoretical calculations, for their values? And, moreover, might it be that this failing does not mean that we need to look for an even deeper theory, but simply reflects that there is no explanation for these observed properties of reality?

One immediate answer to all these questions is yes. As Einstein said some time ago, “The most incomprehensible thing about the universe is that it is comprehensible.”7 The astonishment at our ability to understand the universe at all is easily lost sight of in an age of rapid and impressive progress. However, maybe there is a limit to comprehensibility. Maybe we have to accept that after reaching the deepest possible level of understanding science can offer, there will nevertheless be aspects of the universe that remain unexplained. Maybe we will have to accept that certain features of the universe are the way they are because of happenstance, accident, or divine choice. The success of the scientific method in the past has encouraged us to think that with enough time and effort we can unravel nature’s mysteries. But hitting the absolute limit of scientific explanation—not a technological obstacle or the current but progressing edge of human understanding—would be a singular event, one for which past experience could not prepare us.

Although of great relevance to our quest for the ultimate theory, this is an issue we cannot yet resolve; indeed, the possibility that there are limits to scientific explanation, in the broad way we have stated it, is an issue that may never be resolved. We have seen, for instance, that even the speculative notion of the multiverse, which at first sight appears to present a definite limit to scientific explanation, can be dealt with by dreaming up equally speculative theories that, at least in principle, can restore predictive power.

One highlight emerging from these considerations is the role of cosmology in determining the implications of an ultimate theory. As we have discussed, superstring cosmology is a young field, even by the youthful standards set by string theory itself. It will, undoubtedly, be an area of primary research focus for years to come, and it is likely to be one of the major growth areas of the field. As we continue to gain new insight into the properties of string/M-theory, our ability to assess the cosmological implications of this rich attempt at a unified theory will become ever sharper. It is possible, of course, that such studies